

Signature match: $H(S_{t+2}) = H(S_t) \rightarrow$ repeated state detected

Iteration t : Candidate State S_t

State signature: $H(S_t) = \text{hash}(\text{RTL}, \text{SDC}, \text{UPF})$

Timing: n setup violations ($\text{WNS} < 0$)
DRC: clean

Repair Δ_1
(Buffer Sizing)

Iteration $t + 1$: Candidate State S_{t+1}

State signature: $H(S_{t+1})$, differs from $H(S_t)$

Timing: setup met ($\text{WNS} \geq 0$)
 n routing shorts and hold violations

Repair Δ_2
(Wire Widening)

Iteration $t + 2$: Candidate State S_{t+2}

State signature: $H(S_{t+2}) = H(S_t)$

Timing: n setup violations ($\text{WNS} < 0$)
DRC: clean